Math for CS & AI: Homework 7

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Problem 4.2

Let $(a_n)_{n\geq 1}$ be the sequence that satisfies the recurrence:

$$a_n = \frac{2}{n}\sum_{k=1}^{n-1} a_k + n \quad \text{for all } n \geq 1.$$

(a) Determine a closed-form formula for the generating function of a_n:

$$A(x) = \sum_{n\geq 1} a_n x^n$$

(b) Determine a closed-form formula for a_n. Your answer may involve a Harmonic number.

Answer:

For part (a), we multiply both sides of the recurrence by n:

$$na_n = 2\sum_{k=1}^{n-1} a_k + n^2. \tag{1}$$

Writing the recurrence for $n-1$ gives:

$$(n-1)a_{n-1} = 2\sum_{k=1}^{n-2} a_k + (n-1)^2. \tag{2}$$

Subtracting equation (2) from equation (1) yields:

$$na_n = (n+1)a_{n-1} + 2n - 1. \tag{3}$$

Define $a_0 = 0$. Taking the generating function of both sides, we obtain:

$$\begin{aligned}
\sum_{n\geq 1} na_n x^n &= \sum_{n\geq 1} \left[(n+1)a_{n-1} + 2n - 1\right] x^n, \\
xA'(x) &= x^2 A'(x) + 2xA(x) + \frac{2x}{(1-x)^2} - \frac{x}{1-x}, \\
A'(x) - \frac{2}{1-x}A(x) &= \frac{1+x}{(1-x)^3}.
\end{aligned}$$

Solving the equation:

$$\mu(x) = \exp\left(\int -\frac{2}{1-x}\mathrm{d}x\right) = (1-x)^2,$$
$$(1-x)^2 A(x) = \int \frac{1+x}{1-x}\,dx + C,$$
$$A(x) = \frac{2\ln\left(\frac{1}{1-x}\right) - x + C}{(1-x)^2}.$$

Since $A(0) = 0$, we have $C = 0$. Therefore,

$$A(x) = \frac{2\ln\left(\frac{1}{1-x}\right) - x}{(1-x)^2}.$$

For part (b), let $H(x) = \ln\left(\frac{1}{1-x}\right)$, which is the generating function for the sequence $1/n$. We first compute the coefficient of x^n in $\frac{H(x)}{(1-x)^2}$:

$$[x^n]\frac{H(x)}{(1-x)^2} = \sum_{k=1}^{n} \frac{1}{k} \cdot \binom{n-k+1}{2-1}$$
$$= (n+1)H_n - n,$$

where $H_n = \sum_{k=1}^{n} \frac{1}{k}$ is the harmonic number. Since the coefficient of x^n in $\frac{x}{(1-x)^2}$ is n, we have

$$a_n = 2(n+1)H_n - 3n.$$

Problem 4.3

The Stirling numbers of the second kind $\left\{ {n \atop k} \right\}$ are the number of ways to partition a set of n elements into k non-empty subsets. The order of numbers in each subset is not important, and the order of subsets is also not important. For example, $\left\{ {3 \atop 2} \right\} = 3$ because the set $\{1, 2, 3\}$ can be partitioned into 2 subsets in 3 ways:

$$\{\{1\}, \{2, 3\}\},\ \{\{2\}, \{1, 3\}\},\ \{\{3\}, \{1, 2\}\}.$$

Some other special values are: $\left\{ {n \atop k} \right\} = 0$ if $n < k$, $\left\{ {n \atop n} \right\} = 1$, $\left\{ {n \atop n-1} \right\} = \binom{n}{2}$, $\left\{ {n \atop 2} \right\} = 2^{n-1} - 1$, and $\left\{ {n \atop 1} \right\} = 1$. Fix $k \geq 1$. Determine a closed-form formula for the exponential generating function of $\left\{ {n \atop k} \right\}$:

$$\hat{S}_k(x) = \sum_{n \geq 0} \left\{ {n \atop k} \right\} \frac{x^n}{n!}.$$

Answer:

To derive the exponential generating function, we start with the combinatorial interpretation of Stirling numbers of the second kind. We first consider ordered partitions, i.e., we assign a distinct order to the k subsets. Let a_i denote the number of elements in the i-th subset, where $a_i \geq 1$ for all $1 \leq i \leq k$, we have

$$\left\{ {n \atop k} \right\} = n! \sum_{a_1 + \cdots + a_k = n} \prod_{i=1}^{k} \frac{1}{a_i!}.$$

The generating function for a single non-empty subset is $A(x) = \sum_{k \geq 1} \frac{x^k}{k!} = e^x - 1$. It follows that

$$\hat{S}_k(x) = \frac{A^k(x)}{k!} = \frac{(e^x - 1)^k}{k!}.$$

Problem 4.4

Prove the following identities for all $x \in \mathbb{R}$:

(a) $\sinh(ix) = i \sin x$;

(b) $\cosh(ix) = \cos x$;

(c) $\sinh(x) = -i \sin(ix)$;

(d) $\cosh(x) = \cos(ix)$.

Here, $\sinh(x) = \frac{e^x - e^{-x}}{2}$ and $\cosh(x) = \frac{e^x + e^{-x}}{2}$.

Answer:

For parts (a) and (b), since $e^{ix} = \cos x + i \sin x$, we have

$$\sinh(ix) = \frac{e^{ix} - e^{-ix}}{2} = \frac{\cos x + i \sin x - \cos x + i \sin x}{2} = i \sin x,$$
$$\cosh(ix) = \frac{e^{ix} + e^{-ix}}{2} = \frac{\cos x + i \sin x + \cos x - i \sin x}{2} = \cos x.$$

For parts (c) and (d), since $e^{-x} = e^{i \cdot ix} = \cos(ix) + i \sin(ix)$ and $e^x = e^{i \cdot i(-x)} = \cos(ix) - i \sin(ix)$, we have

$$\sinh(x) = \frac{e^x - e^{-x}}{2} = \frac{\cos(ix) - i \sin(ix) - \cos(ix) - i \sin(ix)}{2} = -i \sin(ix),$$
$$\cosh(x) = \frac{e^x + e^{-x}}{2} = \frac{\cos(ix) - i \sin(ix) + \cos(ix) + i \sin(ix)}{2} = \cos(ix).$$

Problem 4.5

In Example 4.29, we showed that the complex conjugate function $f(z) = \bar{z}$ is continuous everywhere, but it is not differentiable at 0. Prove that f is not differentiable at any point, not just 0.

Answer:

For any point $z_0 = x_0 + iy_0$, we let z approach z_0 along the real axis and the imaginary axis, respectively:

$$\lim_{x \to 0} \frac{f(z_0 + x) - f(z_0)}{(z_0 + x) - z_0} = \lim_{x \to 0} \frac{x}{x} = 1,$$
$$\lim_{y \to 0} \frac{f(z_0 + iy) - f(z_0)}{(z_0 + iy) - z_0} = \lim_{y \to 0} \frac{-iy}{iy} = -1.$$

Therefore, we obtain different limits along different directions of approach, which implies that f is not differentiable at any point z_0.

Problem 4.6

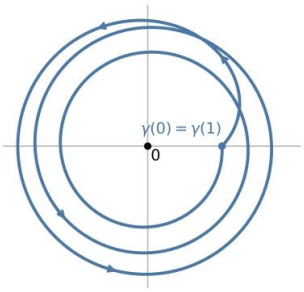
Evaluate the following integrals:

(a) $\int_1^3 (t^2 - i)^2 \mathrm{d}t$;

(b) $\int_0^{\pi/4} e^{-2it} \mathrm{d}t$;

(c) Let $\gamma : [0, 1] \to \mathbb{C}$ be a closed curve defined by $\gamma(t) = (1+3t(1-t))e^{i(t-t^2+6t^3)\pi}$ (note that $\gamma(0) = \gamma(1)$). Compute:

$$\int_\gamma (z^{-2} + z^{-1} + 1 + z)\mathrm{d}z.$$



Answer:

For part (a), we have

$$\begin{aligned}\int_1^3 (t^2 - i)^2 \mathrm{d}t &= \int_1^3 (t^4 - 1 - 2it^2)\mathrm{d}t \\ &= \int_1^3 (t^4 - 1)\mathrm{d}t - 2i\int_1^3 t^2 \mathrm{d}t \\ &= \frac{232}{5} - \frac{52}{3}i.\end{aligned}$$

For part (b), we have

$$\begin{aligned}\int_0^{\pi/4} e^{-2it} \mathrm{d}t &= \int_0^{\pi/4} (\cos(-2t) + i\sin(-2t))\mathrm{d}t \\ &= \frac{1}{2}\int_0^{\pi/2} \cos t \mathrm{d}t - \frac{i}{2}\int_0^{\pi/2} \sin t \mathrm{d}t \\ &= \frac{1}{2} - \frac{1}{2}i.\end{aligned}$$

For part (c), we derived in Examples 4.47 to 4.49 that the integral of z^n along any closed curve not passing through the origin is 0 for all integers $n \neq -1$, and the integral for z^{-1} is $2\pi i$ times the number of times γ winds around the origin, counted counterclockwise. Only the z^{-1} term contributes to the integral. From the given figure, we observe that

$$\int_\gamma (z^{-2} + z^{-1} + 1 + z)\mathrm{d}z = 6\pi i.$$

Problem 4.7

Show that the following variant of the identity theorem (Theorem 4.61) is true, or give a counterexample. Let $f : S_1 \to \mathbb{C}$ and $g : S_2 \to \mathbb{C}$ be holomorphic functions on two connected open sets S_1 and S_2. If there exists a non-empty open set $U \subseteq S_1 \cap S_2$ such that $f(z) = g(z)$ for every $z \in U$, then $f(z) = g(z)$ for every $z \in S_1 \cap S_2$.

For this problem, you do not have to be fully rigorous. Just state the result and give a brief explanation.

Answer:

This variant of the identity theorem is false. Counterexamples exist when the intersection $S_1 \cap S_2$ is not connected, as we can have $f(z) = g(z)$ on some connected components of the intersection but not on others.

To construct a concrete counterexample, let S_1 be a horizontal hollow rectangular annulus, and let S_2 be a vertical hollow rectangular annulus of the same shape. This construction ensures that both S_1 and S_2 are connected individually, but their intersection $S_1 \cap S_2$ consists of at least two disjoint open sets. We can then define f and g to be equal on one of these open sets (the U in the problem statement) but different on the other, which violates the conclusion of the proposed variant.

Problem 4.8

Determine the radius of convergence of the Taylor series of each of the following functions around 0:

(a) $f(z) = \frac{1}{(z-5)(z^2+4)}$;

(b) $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$;

(c) $\text{softplus}(z) = \text{Log}(1 + e^z)$.

Answer:

For part (a), the non-removable singularities of $f(z)$ are 5, $2i$, and $-2i$. The closest singularity to the origin is at distance 2, so the radius of convergence of the Taylor series of f around 0 is 2.

For part (b), the non-removable singularities of $\tanh(z)$ are $\frac{(2k+1)i\pi}{2}$ for all integers k. The closest singularities to the origin are $\pm\frac{i\pi}{2}$, at distance $\frac{\pi}{2}$, so the radius of convergence of the Taylor series of tanh around 0 is $\frac{\pi}{2}$.

For part (c), the non-removable singularities of $\text{softplus}(z)$ are $(2k+1)i\pi$ for all integers k. The closest singularities to the origin are $\pm i\pi$, at distance π, so the radius of convergence of the Taylor series of softplus around 0 is π.

Problem 4.9

The error function $\operatorname{erf} : \mathbb{R} \to \mathbb{R}$ is defined as:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \mathrm{d}t$$

Find a power series $f(z) = \sum_{n\geq 0} a_n z^n$ such that the series converges absolutely for all $z \in \mathbb{C}$ and $f(x) = \operatorname{erf}(x)$ for all $x \in \mathbb{R}$.

Answer:

By the definition of the error function, for all $x \in \mathbb{R}$, we have

$$\begin{aligned}
\operatorname{erf}(x) &= \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \mathrm{d}t \\
&= \frac{2}{\sqrt{\pi}} \int_0^x \sum_{n\geq 0} \frac{(-1)^n}{n!} t^{2n} \mathrm{d}t \\
&= \frac{2}{\sqrt{\pi}} \sum_{n\geq 0} \frac{(-1)^n}{n!} \int_0^x t^{2n} \mathrm{d}t \\
&= \frac{2}{\sqrt{\pi}} \sum_{n\geq 0} \frac{(-1)^n}{n!} \frac{x^{2n+1}}{2n+1}.
\end{aligned}$$

Note that all even-indexed coefficients are zero, and for odd indices we have $a_{2n+1} = \frac{2\cdot(-1)^n}{\sqrt{\pi}\cdot n!\cdot(2n+1)}$. We compute $\limsup_{n\to\infty} |a_n|^{1/n} = 0$ since the factorial term in the denominator dominates all polynomial growth. By the Cauchy–Hadamard theorem, this power series converges absolutely for all $z \in \mathbb{C}$. Therefore, the desired entire function is:

$$f(z) = \frac{2}{\sqrt{\pi}} \sum_{n\geq 0} \frac{(-1)^n z^{2n+1}}{n!(2n+1)}.$$

Problem 4.10

In this problem, we will guide you through the proof of Laurent's theorem (Theorem 4.65). Let f be holomorphic on a punctured disk $D_r(z_0) \setminus \{z_0\}$ for some $z_0 \in \mathbb{C}$ and some $r > 0$. Fix $z \in D_r(z_0) \setminus \{z_0\}$. Let $\epsilon > 0$ be an arbitrary positive number such that $z_0 \notin D_\epsilon(z)$.

(a) Prove the following identity:

$$f(z) = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(w)}{w - z} \mathrm{d}w - \frac{1}{2\pi i} \int_{C_\epsilon(z_0)} \frac{f(w)}{w - z} \mathrm{d}w.$$

(b) Show that for any $w \in D_\epsilon(z_0)$, we have:

$$\frac{f(w)}{w - z} = -\sum_{n \geq 1} f(w)(w - z_0)^{n-1}(z - z_0)^{-n}.$$

(c) Let $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 1}$ be the following sequences:

$$a_n = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(w)}{(w - z_0)^{n+1}} \mathrm{d}w, \quad b_n = \frac{1}{2\pi i} \int_{C_\epsilon(z_0)} f(w)(w - z_0)^{n-1} \mathrm{d}w.$$

Prove that:

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} b_n (z - z_0)^{-n}.$$

(d) Prove that if $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$ for some sequence $(a_n)_{n \in \mathbb{Z}}$, then for all $n \in \mathbb{Z}$:

$$a_n = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(w)}{(w - z_0)^{n+1}} \mathrm{d}w$$

As the right-hand side only depends on f and z_0, this automatically proves the uniqueness of the coefficients in the Laurent series.

For this problem, you may freely swap the order of integration and summation, without providing any justification.

Answer:

For part (a), we follow the argument in the proof of Theorem 4.57 (Cauchy's integral formula). We choose $r' > 0$ such that $C_{r'}(z)$ lies entirely inside the annulus between $C_\epsilon(z_0)$ and $C_r(z_0)$, and is disjoint from both curves. As shown in Figure 1 below, we have:

$$\int_{C_r(z_0)} \frac{f(w)}{w - z} \mathrm{d}w - \int_{C_\epsilon(z_0)} \frac{f(w)}{w - z} \mathrm{d}w - \int_{C_{r'}(z)} \frac{f(w)}{w - z} \mathrm{d}w = 0$$

since the integrand $\frac{f(w)}{w-z}$ is holomorphic in the region bounded by these three curves (both singularities $w = z_0$ and $w = z$ lie inside the inner circles, not in the region itself). Then we apply Cauchy's integral formula to $C_{r'}(z)$ and obtain:

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_{C_{r'}(z)} \frac{f(w)}{w - z} \mathrm{d}w \\ &= \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(w)}{w - z} \mathrm{d}w - \frac{1}{2\pi i} \int_{C_\epsilon(z_0)} \frac{f(w)}{w - z} \mathrm{d}w. \end{aligned}$$

Problem 4.10 continued on next page...

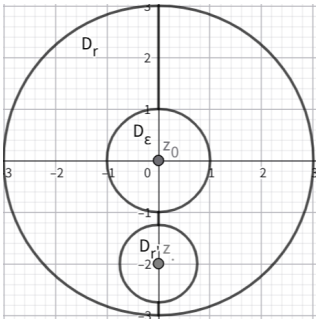


Figure 1: Part (a).

For part (b), we expand the integrand as a geometric series. This series converges absolutely because:

$$\left|\frac{w-z_0}{z-z_0}\right| = \frac{|w-z_0|}{|z-z_0|} < 1.$$

The expansion is:

$$\begin{aligned}\frac{f(w)}{w-z} &= -\frac{f(w)}{z-z_0}\cdot\frac{1}{1-\frac{w-z_0}{z-z_0}} \\ &= -\frac{f(w)}{z-z_0}\sum_{n\geq 0}\frac{(w-z_0)^n}{(z-z_0)^n} \\ &= -\sum_{n\geq 1} f(w)(w-z_0)^{n-1}(z-z_0)^{-n}.\end{aligned}$$

For part (c), combining the results from parts (a) and (b), and substituting the geometric series expansion into the integral, we get:

$$\begin{aligned}f(z) &= \frac{1}{2\pi i}\int_{C_r(z_0)}\frac{f(w)}{w-z}\mathrm{d}w - \frac{1}{2\pi i}\int_{C_\epsilon(z_0)}\frac{f(w)}{w-z}\mathrm{d}w \\ &= \sum_{n\geq 0}\left(\frac{1}{2\pi i}\int_{C_r(z_0)}\frac{f(w)}{(w-z_0)^{n+1}}\mathrm{d}w\right)(z-z_0)^n + \sum_{n\geq 1}\left(\frac{1}{2\pi i}\int_{C_\epsilon(z_0)} f(w)(w-z_0)^{n-1}\mathrm{d}w\right)(z-z_0)^{-n} \\ &= \sum_{n=0}^{\infty} a_n(z-z_0)^n + \sum_{n=1}^{\infty} b_n(z-z_0)^{-n}.\end{aligned}$$

For part (d), since the only singularity of $\frac{f(w)}{(w-z_0)^{n+1}}$ is z_0, by the integral path deformation invariance, the integral along $C_\epsilon(z_0)$ equals the integral along $C_r(z_0)$. Therefore,

$$\begin{aligned}f(z) &= \sum_{n\geq 0}\left(\frac{1}{2\pi i}\int_{C_r(z_0)}\frac{f(w)}{(w-z_0)^{n+1}}\mathrm{d}w\right)(z-z_0)^n + \sum_{n\geq 1}\left(\frac{1}{2\pi i}\int_{C_\epsilon(z_0)} f(w)(w-z_0)^{n-1}\mathrm{d}w\right)(z-z_0)^{-n} \\ &= \sum_{n\geq 0}\left(\frac{1}{2\pi i}\int_{C_r(z_0)}\frac{f(w)}{(w-z_0)^{n+1}}\mathrm{d}w\right)(z-z_0)^n + \sum_{n\leq -1}\left(\frac{1}{2\pi i}\int_{C_r(z_0)}\frac{f(w)}{(w-z_0)^{n+1}}\mathrm{d}w\right)(z-z_0)^n \\ &= \sum_{n=-\infty}^{\infty}\left(\frac{1}{2\pi i}\int_{C_r(z_0)}\frac{f(w)}{(w-z_0)^{n+1}}\mathrm{d}w\right)(z-z_0)^n.\end{aligned}$$